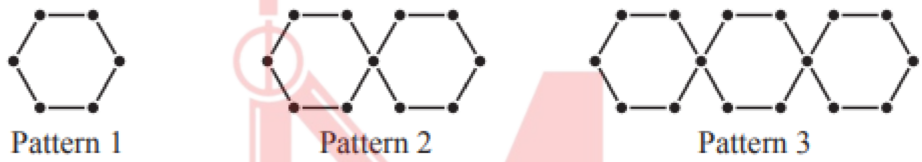
Number Sequences

Worksheet 1 (Non - Calculator)

Arithmetic Sequence Only

1. W15/qp11/q18

A sequence of patterns is made using dots and lines



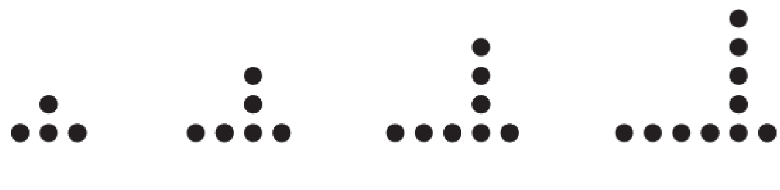
Pattern number (p)	1	2	3	4
Number of dots (d)	6	11	16	

a. Complete the table for Pattern 4. [1]

b. Find a formula for the number of dots, d, in Pattern p. [2]

2. S16/qp11/q17

A sequence of diagrams is made using counters.



a. Complete the table.

Diagram number	1	2	3	4	5
Number of counters	4	6	8		

[1]

b. Find an expression, in terms of n, for the number of counters in Diagram n [1]

c. In this sequence, Diagram p has 200 counters. Find the value of p. [2]

3. W16/qp12/q26

Two sequences have 1, 3, 5 as their first three terms.

a. In the first sequence, each term is 2 more than the term before it.

   i. Find an expression, in terms of n, for the nth term. [1]

   ii. The k^{th} term of this sequence is 841. Find the value of k. [1]

4. W18/qp11/q7

The value of each term of a sequence is 4 more than the value of the term before it. The third term is 17 and the fourth term is 21.

a. Find the first term. [1]

b. Find an expression for the nth term of this sequence. Give your answer in its simplest form. [2]

5. W18/qp12/q11

Here are the first five terms of a sequence.

$$\frac{3}{4} \quad \frac{7}{8} \quad \frac{11}{12} \quad \frac{15}{16} \quad \frac{19}{20}$$

a. Write down the next two terms. [1]

b. The kth term is $\frac{1199}{1200}$. Find k. [1]

c. Find an expression, in terms of n, for the nth term. [2]

6. S19/qp11/q12

The r^{th} term of a sequence, u_r , is given by $u_r = 3r + 2$.

a. Find the third term, u_3, in this sequence. [1]

b. Given that $u_k = 50$, find the value of k. [1]

7. S19/qp12/q22

Here are the first three patterns in a sequence made using dots and lines.

Pattern 1 Pattern 2 Pattern 3

Pattern number	1	2	3	4	5
Number of dots	3	6			
Number of lines	2	7			

a. Complete the table for the first five patterns in this sequence. [2]

b. Find an expression, in terms of n, for the number of lines in Pattern n. [2]

c. Anwar makes one of these patterns using 92 lines. Find the number of dots in Anwar's pattern. [2]

8. S20/qp11/q3

The numbers in this sequence increase by the same amount each time.

....   1.4   2.3   3.2   ....

Fill in the missing numbers. [2]

9. S20/qp11/q23

a. The formula for the nth term of a sequence is $2n^3$. Find the 3rd term of this sequence. [1]

b. Here are the first four terms of another sequence.

$$\frac{4}{3} \qquad \frac{9}{5} \qquad \frac{16}{7} \qquad \frac{25}{9}$$

i. Write down the next term of this sequence. [1]

ii. Find a formula for the nth term of this sequence. [3]

10. S20/qp12/q17

Here are the first four terms of a number sequence

$$T_1 = 1^2 + 3 = 4$$

$$T_2 = 2^2 + 8 = 12$$

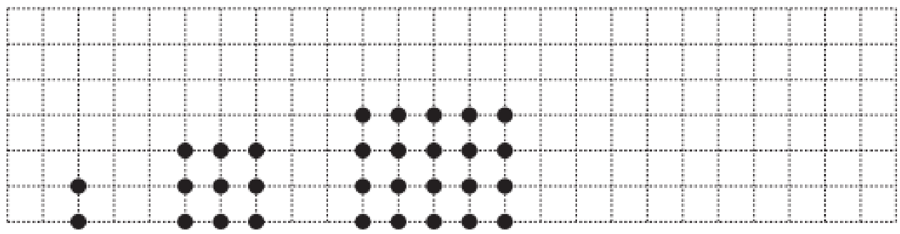
$$T_3 = 3^2 + 13 = 22$$

$$T_4 = 4^2 + 18 = 34$$

a. Find T_5 . [1]

b. Find an expression, in terms of n, for T_n . [3]

11. W20/qp12/q20



Pattern 1 Pattern 2 Pattern 3 Pattern 4

The diagram shows a sequence of patterns. Each pattern has one more row, and two more dots in each row, than the pattern before it.

a. On the diagram, draw Pattern 4. [1]

b.

i. Complete the table for the first four patterns in this sequence. [1]

Pattern number	1	2	3	4		n
Number of rows	2	3	4			p
Number of dots **in each row**	1	3				q
Total number of dots	2	9				

ii. Find an expression, in terms of n, for p. [1]

iii. Find an expression, in terms of n, for q. [1]

Answer Key

1. W15/qp11/q18

18 (a)	21	1	
(b)	$5p + 1$ oe	2	**C1** for $5p + c$; or for $kp + 1$ $(k \neq 0)$

2. S16/qp11/q17

17 (a)	10, 12	1	
(b)	$2n + 2$	1	
(c)	99	2 *	**M1** for *their* (b) = 200

3. W16/qp12/q26

26 (a) (i)	$2n - 1$ oe	1	
(ii)	421	1	

4. W18/qp11/q7

7(a)	9	**1**	
7(b)	$4n + 5$ cao	**2**	**B1** for $4n + k$ oe

5. W18/qp12/q11

11(a)	$\frac{23}{24}$ $\frac{27}{28}$	**1**	
11(b)	300	**1**	
11(c)	$\frac{4n-1}{4n}$ oe	**2**	**B1** for $\frac{\dots}{4n}$, or for $4n - 1$ oe

6. S19/qp11/q12

12(a)	11	**1**	
12(b)	16	**1**	

7. S19/qp12/q22

22(a)	9 12 15 12 17 22	**2**	**B1** for one row correct
22(b)	$5n - 3$ oe final answer	**2**	**B1** for $5n + k$ oe seen
22(c)	57	**2**	**M1** for *their* $(5n - 3) = 92$ or **B1** for $n = 19$ soi or for answer 19

8. S20/qp11/q3

3	(0).5 4.1	**2**	**B1** for each or **M1** for difference = (0).9 soi

9. S20/qp11/q23

23(a)	54	**1**	
23(b)(i)	$\frac{36}{11}$	**1**	
23(b)(ii)	$\frac{(n+1)^2}{2n+1}$ oe	**3**	**B2** for $2n + 1$ oe OR **B1** for $(n + 1)^2$ oe **B1** for $2n$ associated with 3, 5 ,7, 9

10. S20/qp12/q17

17(a)	48	**1**	
17(b)	$n^2 + 5n - 2$ oe final answer	**3**	**B2** for answer $n^2 + 5n + k$ oe or for $5n - 2$ oe seen or **B1** for answer $n^2 + an + b$ or for $5n + k$ oe seen

11. W20/qp12/q20

20(a)	Rectangle with base 7 dots, height 5 dots	**1**	
20(b)(i)	3 4 4 5 5 7 20 35	**1**	
20(b)(ii)	$n + 1$ oe final answer	**1**	
20(b)(iii)	$2n - 1$ oe final answer	**1**	